Networks and Security

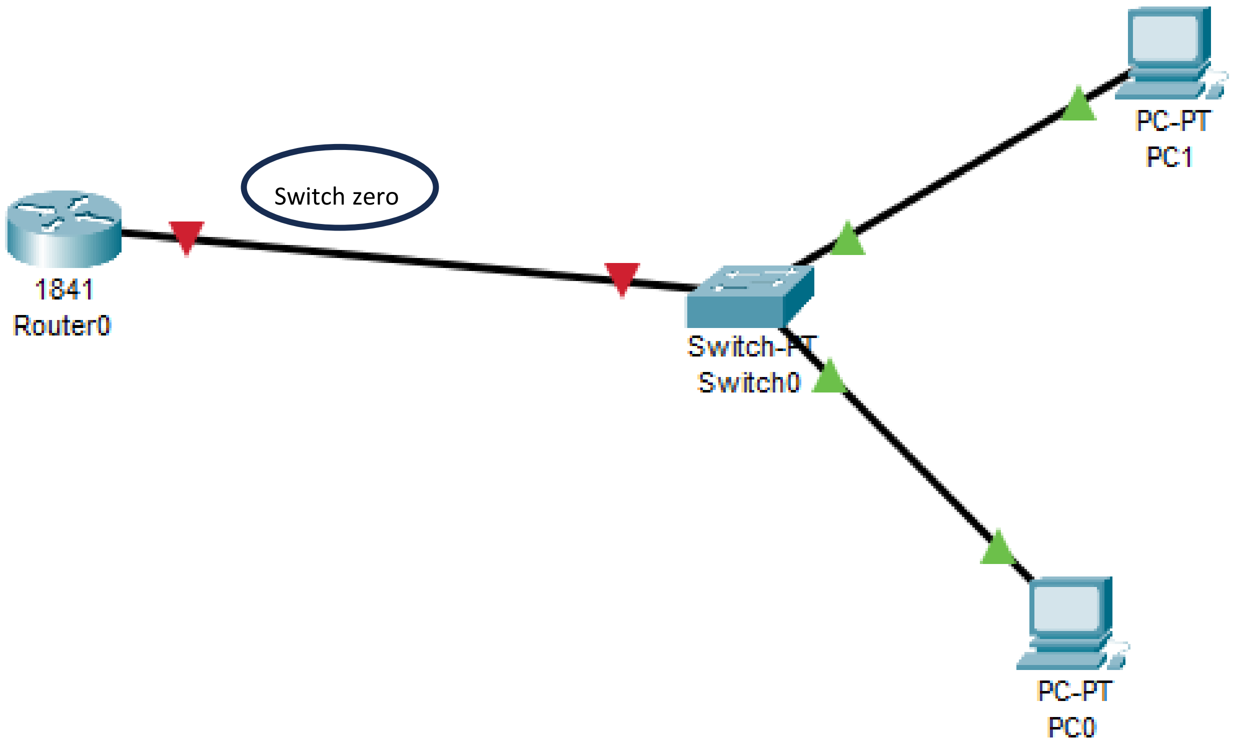
Alireza Esfahani *BSc, MSc, PhD, FHEA, M IEEE, M ECSO*
Senior Lecturer in Cyber Security
Cyber Security Course Leader
School of Engineering and Computing
University of West London

UNIVERSITY OF WEST LONDON
The Career University

DHCP Configuration

Set up your network

- Router (1841)
- Switch (PT-Switch)
- 2 PCs



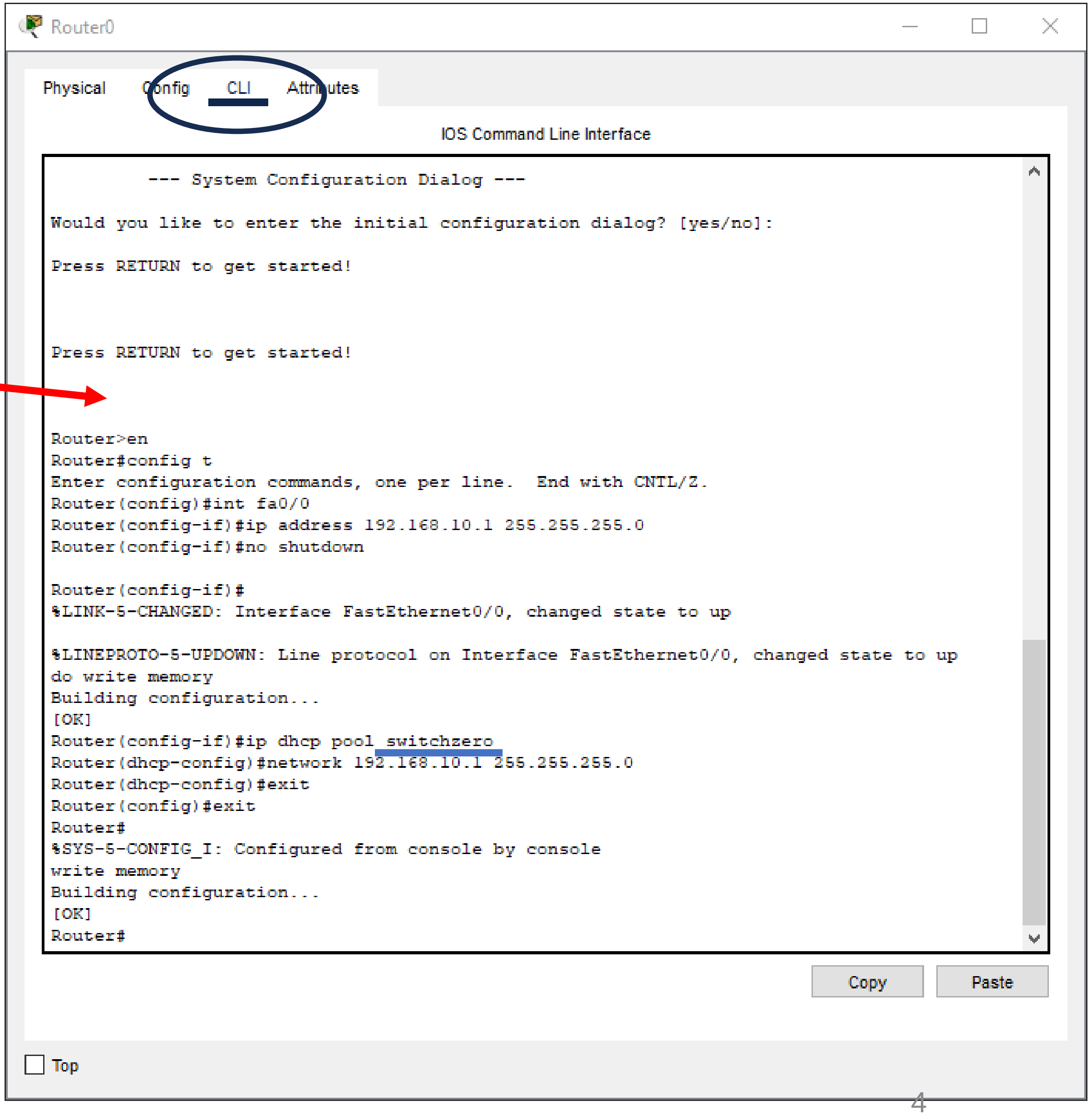
Router

On your router and from CLI, follow the instruction provided through the CLI window:

Router>en

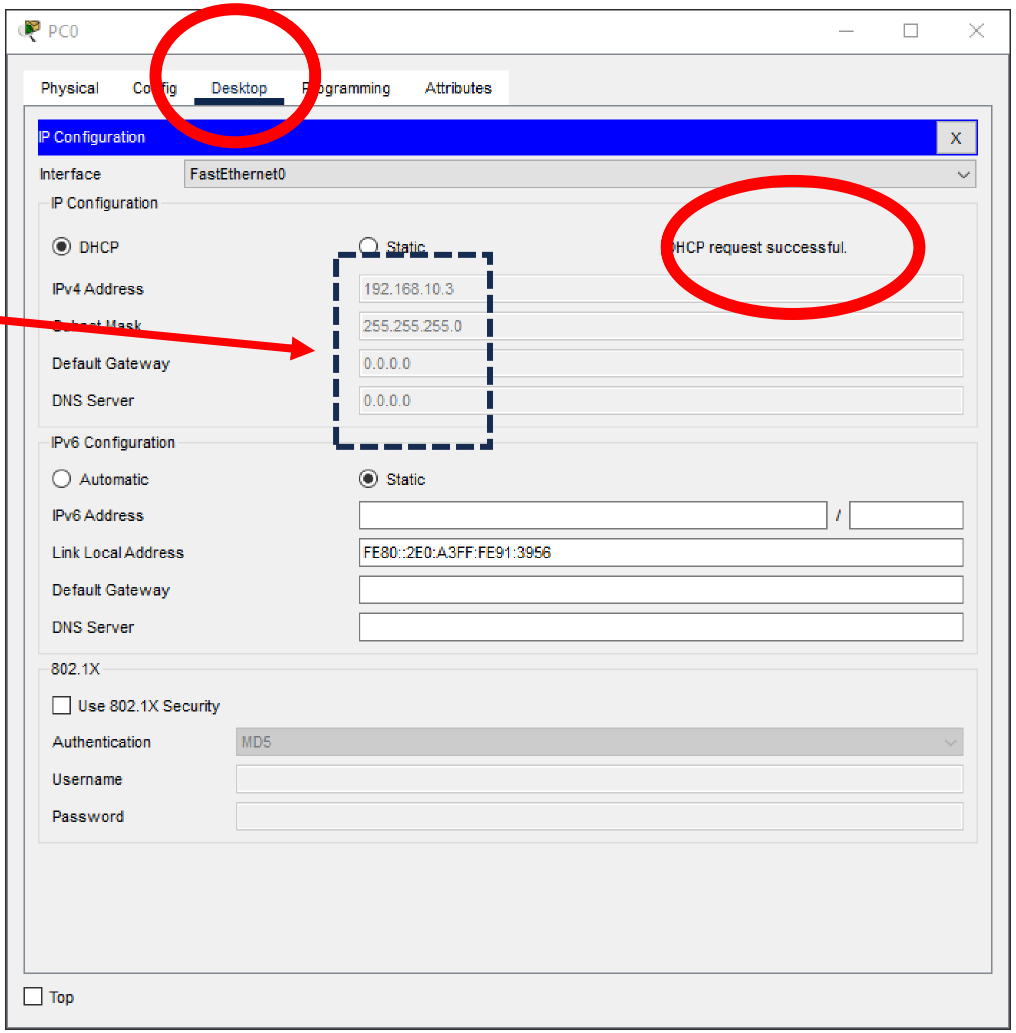
Router#config t

Continue…..



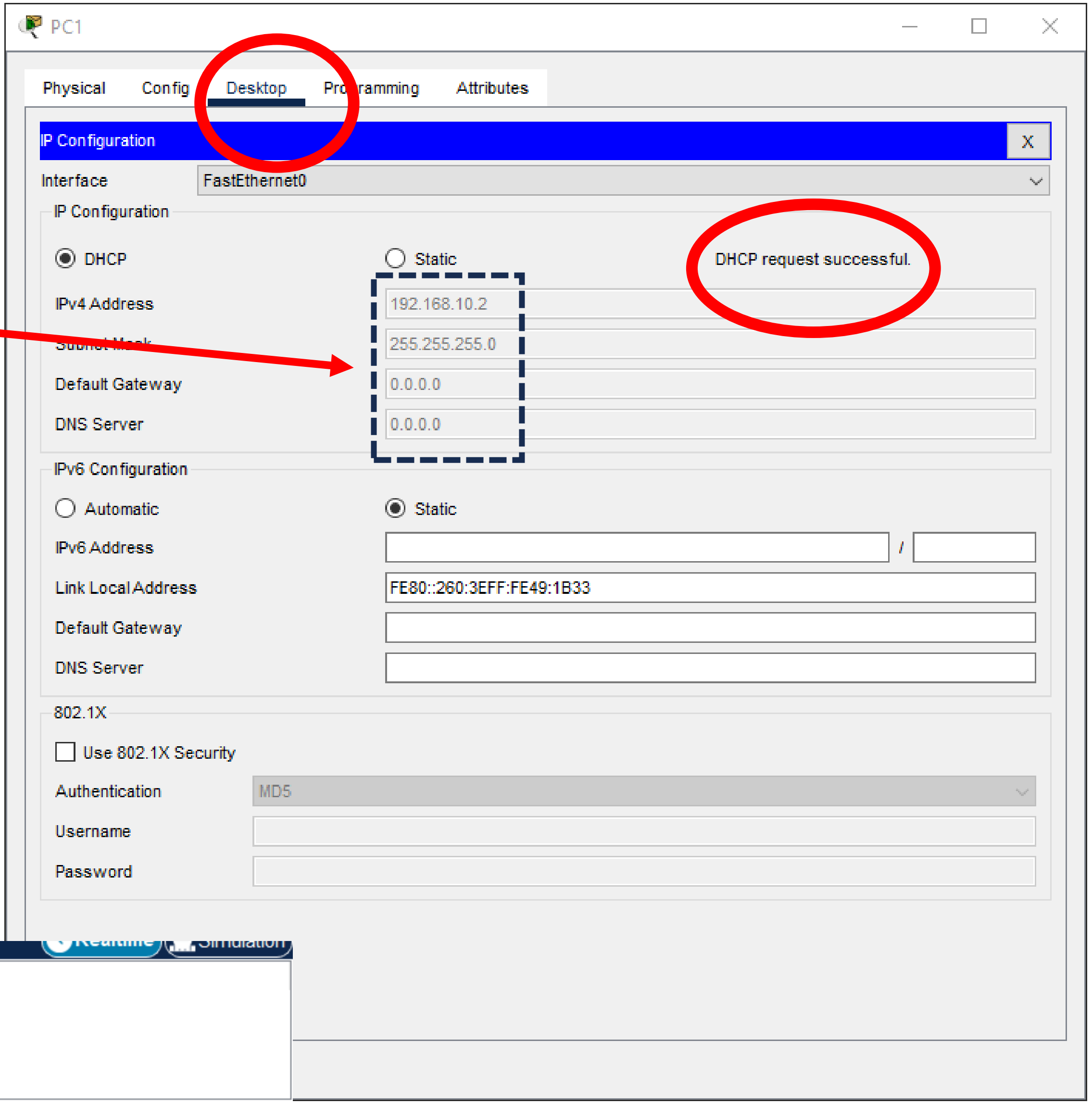
PC0

On PC0 and from Desktop tab, click on IP Configuration, and select DHCP.

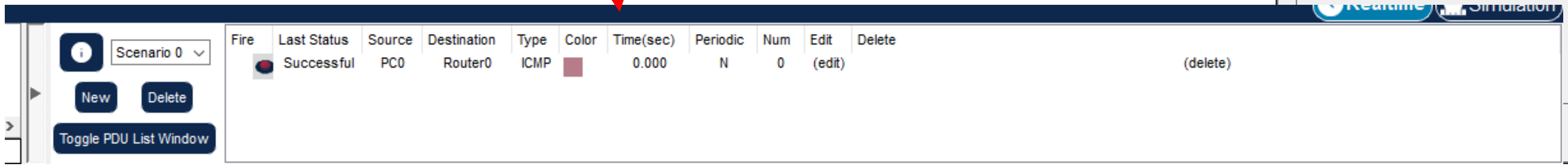


PC1

On PC1 and from Desktop tab, click on IP Configuration, and select DHCP.



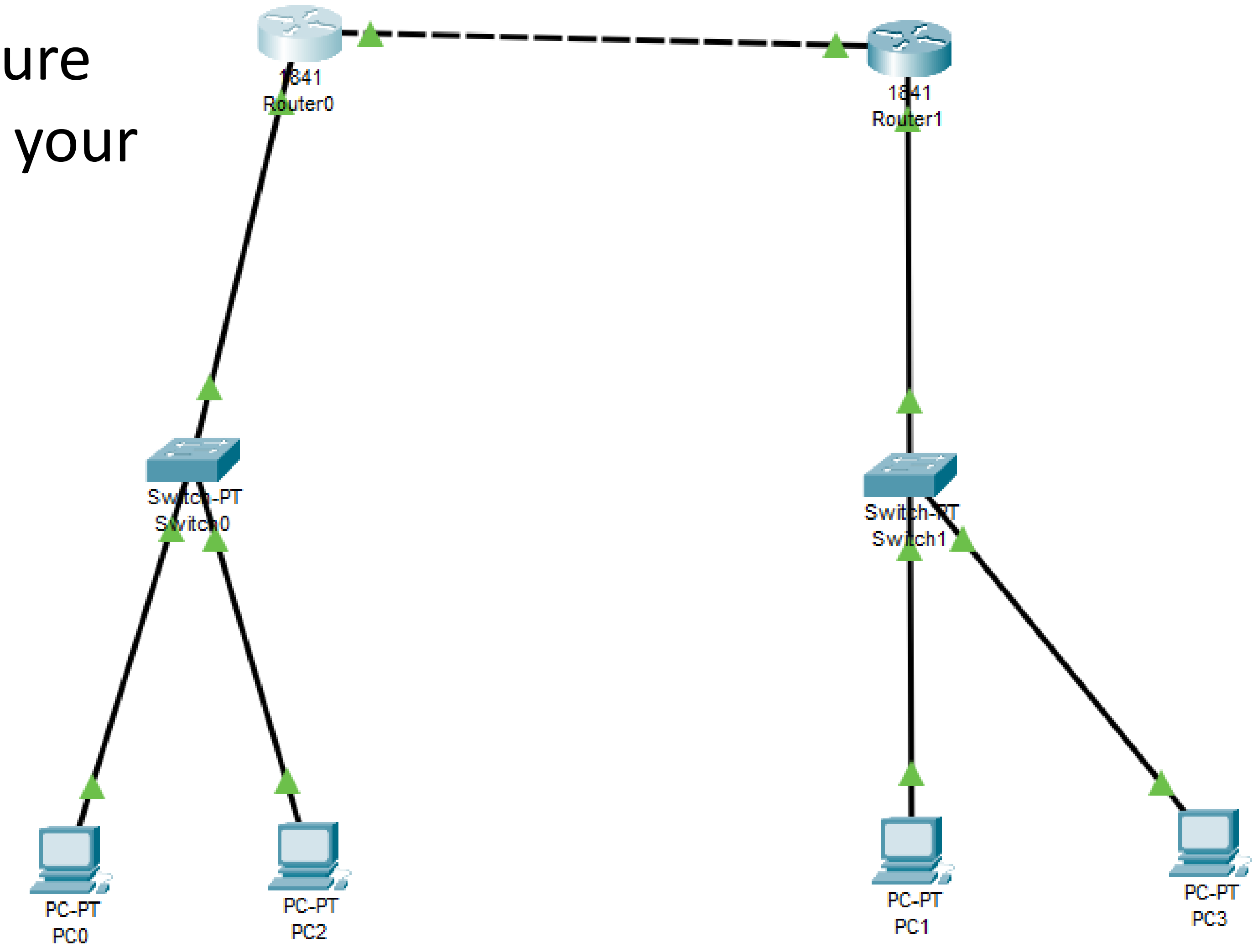
Finally, check the connectivity between PCs and the router.



Exercise

Objectives

Design the given topology and configure DHCP to demonstrate that the PCs in your network are successfully connected.



Hints

You may need to use some useful commands to set up and verify the network, such as those for configuring interfaces, and enabling DHCP. Below are some commands that might help, but you may need to do some additional research:

- *For instance, on Router 0 (R0):*
- *Exclude the router's IP address from the DHCP pool:*

- *R0(config)# ip dhcp excluded-address 192.168.10.1*
- *R0(config)# ip dhcp pool LAN1*
- *R0(dhcp-config)# network 192.168.10.0 255.255.255.0*
- *R0(dhcp-config)# default-router 192.168.10.1*

- *On Router 0 (R0):*
- *Add a static route to reach PC2's network via Router 1:*

- *R0(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2*